

# From Particles to Waves

Towards Coupling Dynamic  
Snow Modelling and Spectral  
Elements to Detect  
Avalanche Breakoff in DAS  
Data

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# Why Model Crack Propagation in Snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

# Relevance

- Crack propagation at different speeds: sub-Rayleigh (below  $v_s$  of material) and Supershear (above  $v_s$ ) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation  $\rightarrow$  critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

# Overview

- **Goal:** Couple dynamic snow modelling with SE to model crack propagation in snow
- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations



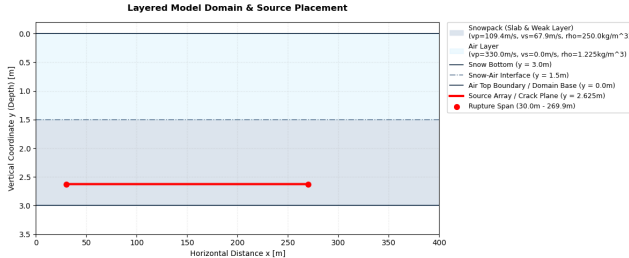
# Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

# Research Questions

- What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

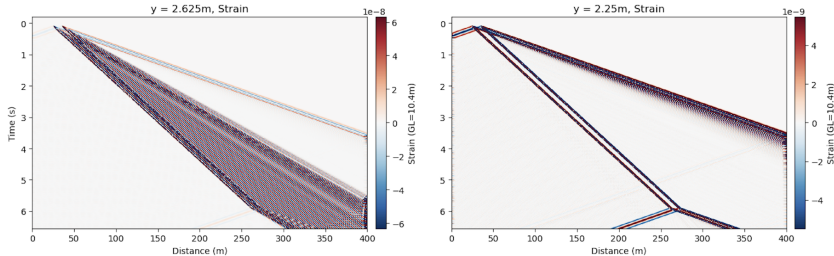
# Simple Model I



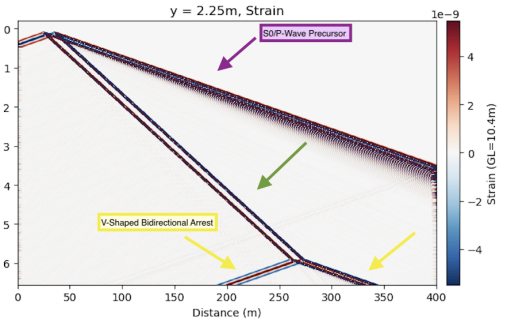
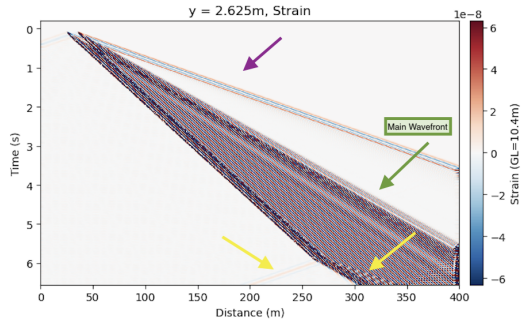
**Figure:** Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red. Propagation speeds according to Trottet et al., 2022.

- SALVUS Output:  
Displacement field  $u$ , used to simulate DAS using Pyber library by Noe, 2025
- $\frac{\partial u_i}{\partial t} \rightarrow$  velocity in  $x$  and  $y$
- $\frac{\partial u_i}{\partial j} \rightarrow$  strain in  $x$  and  $y$  ( $\epsilon$ )
- $\frac{\partial^2 u_i}{\partial t \partial j} \rightarrow$  strain rate ( $\dot{\epsilon}$ )

# Simple Sub-Rayleigh: Strain

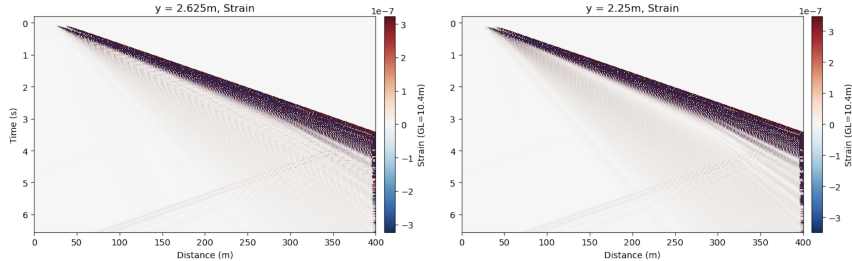


**Figure:** Horizontal Strain for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

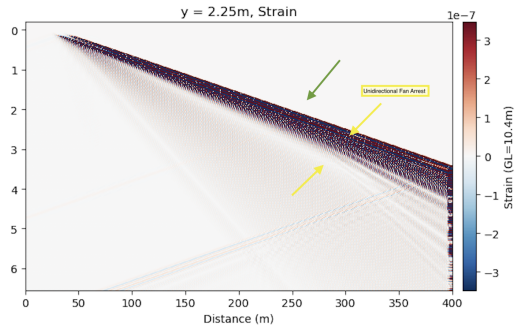
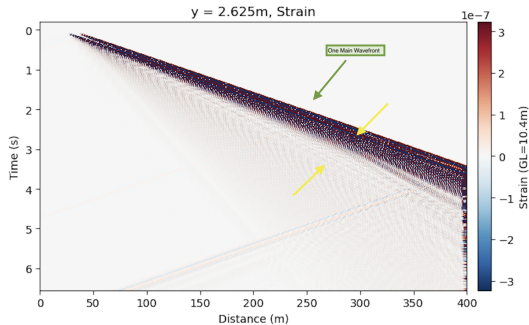


- Precursor emitted by all sources but they interfere (Madariaga, 1976), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit ("Waves in Plates", 2014)
- S0/P-wave precursor observed in field data by Herwijnen and Schweizer, 2011
- Oscillatory behavior of both wavefronts (polarity flip at adjacent timesteps)
- Significant weakening of amplitude between two depths

# Simple Supershear: Strain



**Figure:** Horizontal Strain for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.



- Weakening and broadening of main wavefront (Aki & Richards, 1981)
- Oscillatory wavefront
- At  $x=270\text{m}$ , unidirectional fan-like arrest front: rupture outruns vs, so arrest energy is only propagated forward (Madariaga, 1977)
- Supershear larger amplitude compared to sub-Rayleigh

## Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and antisymmetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (Achenbach, 1973; Lamb, 1917; "Waves in Plates", 2014), so in this case yy
- Confirmed by animations of wavefield:



## Simple model: Lamb Wave Identification II

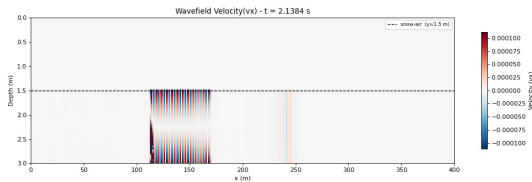


Figure: Animation still for Sub-Rayleigh Vx

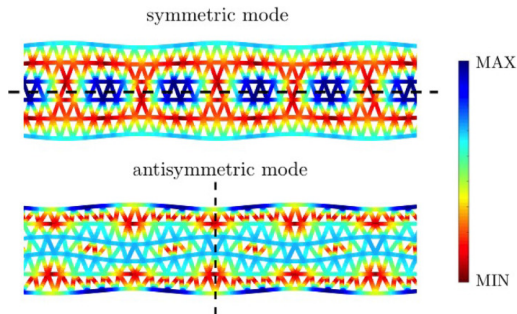
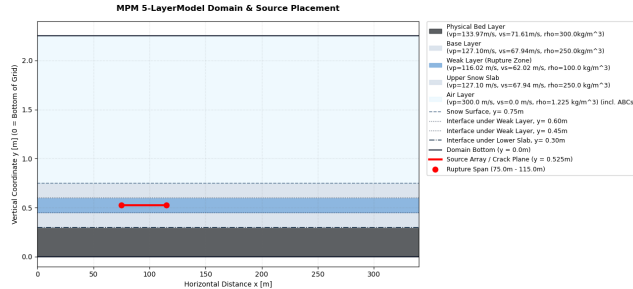


Figure: Lamb wave modes intensity, taken from Carta et al., 2023

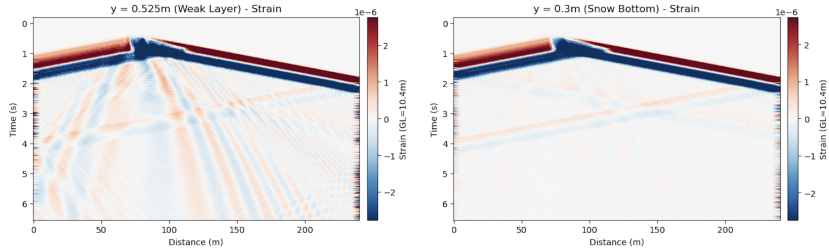
# MPM-Guided Model I



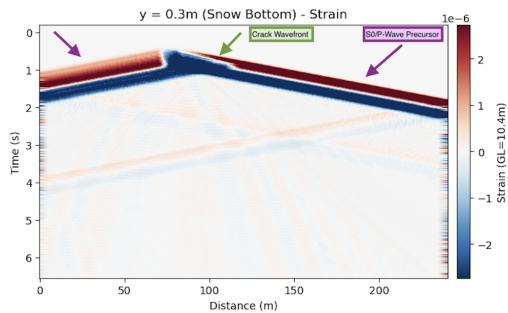
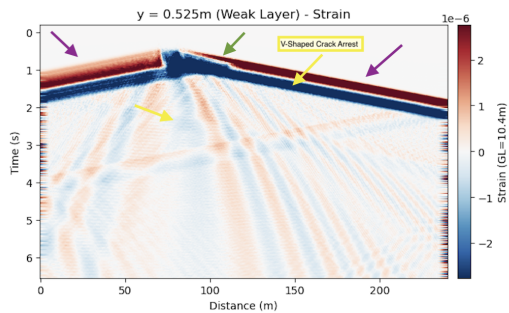
**Figure:** Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

# MPM-Guided: Strain



**Figure:** Horizontal Strain for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.



# Anticrack in MPM

- Static field represents permanent deformation after sources have fired (Archambeau, 1968; Zhu & Rivera, 2002)
- MPM STFS are not symmetric: contrary to Ricker wavelets, they do not integrate to 0 over time → net displacement of material (Aki & Richards, 1981; Zhu & Rivera, 2002)
- Cumulative integral of MPM STFs is negative: collapse of the weak layer, as observed by (Gaume et al., 2018; Heierli et al., 2008; Heierli et al., 2003; Trottet et al., 2022)
- All components of MPM STF (directionality) indicate mixed-mode anticrack instead of pure opening or shearing (mode I, II or III)

# MPM-Guided Model: Lamb Wave Identification

- Like in simple model: oscillatory wavefronts  $\rightarrow$  A0 mode

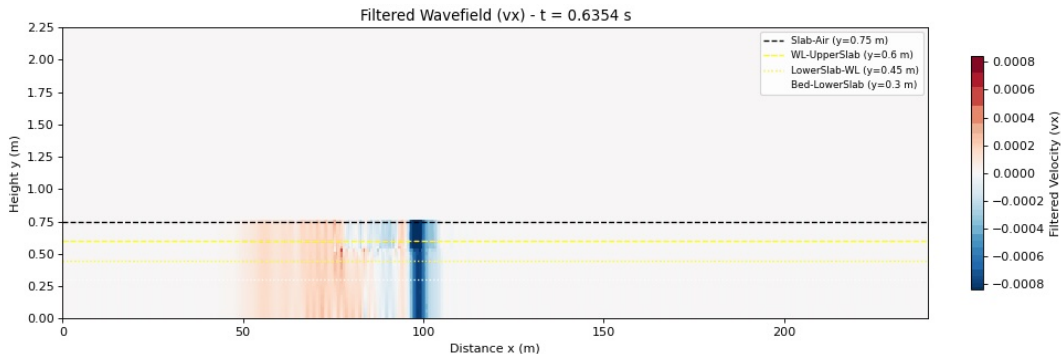
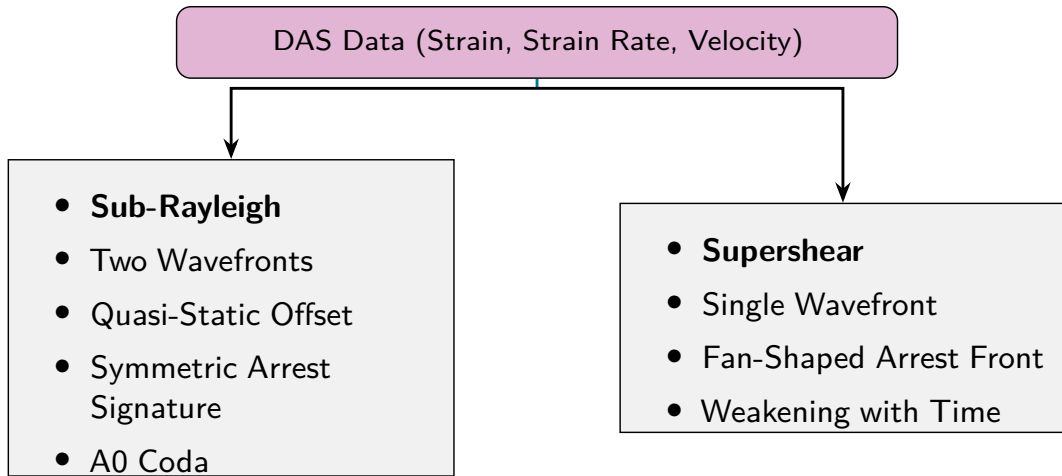


Figure: Animation still for MPM  $V_x$

# Summary of Results & Discussion I

- S0 Precursor observed in all Strain & Strain Rate Plots, as well as in traces
- Main wavfront in all cases is A0  $\rightarrow$  source normal to slab midplane
- Different rupture arrest structures depending on propagation speed
- Quasi-static wavefield for sources that do not integrate to 0 over time
- Weakening of amplitude only in simple case  $\rightarrow$  also thicker slab

## Implications for DAS field data





# Outlook

- Add topography to models
- Verify Supershear effects for more complex STFs as well
- Inhomogeneous, multi-layered snowpack
- Transitional propagation regimes
- Comparison to field data → amplitude of signals compared to noise
- 3D model

## Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

# Questions?

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# MPM STF Workflow

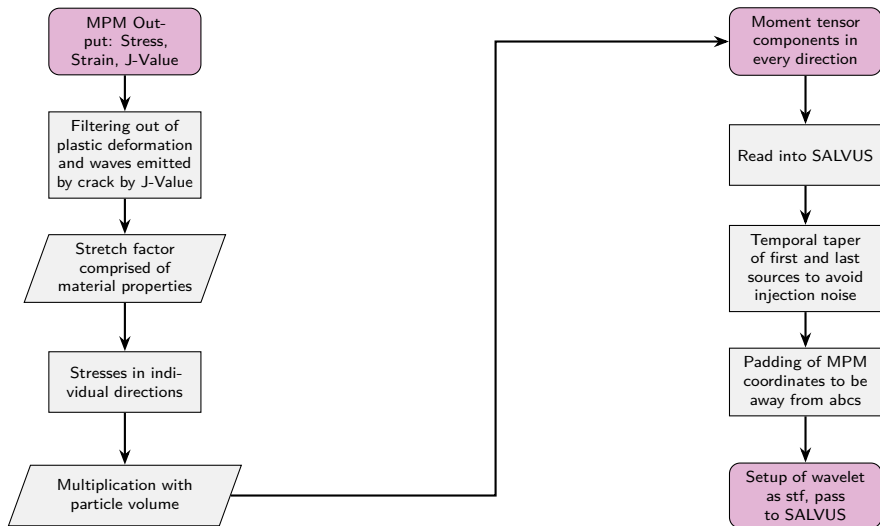
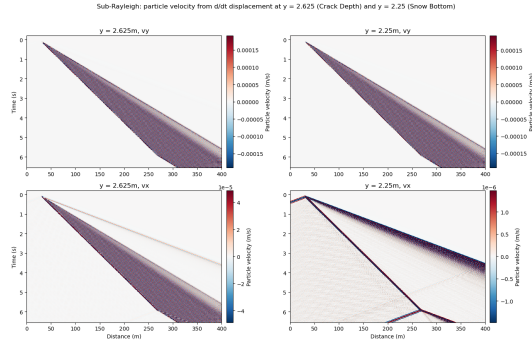


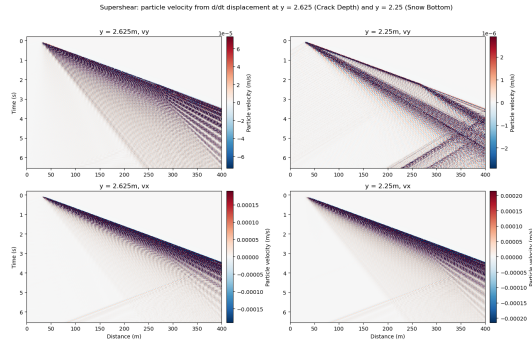
Figure: Workflow for conversion of MPM data and input as SALVUS STF.

# Simple model velocity plots I



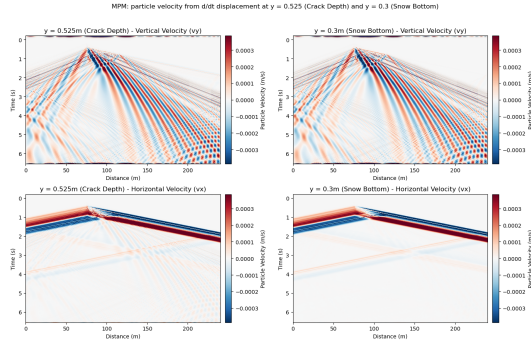
**Figure:** Simple model Sub-Rayleigh velocity plots. Crack depth on left side, snow bottom on right.  $v_y$  at top,  $v_x$  at bottom. Velocity derived from displacement.

# Simple model velocity plots II



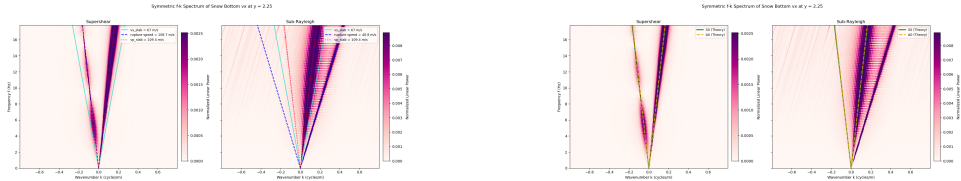
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# MPM velocity plots



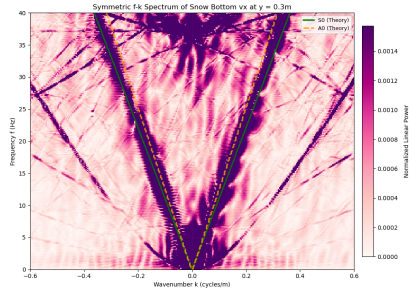
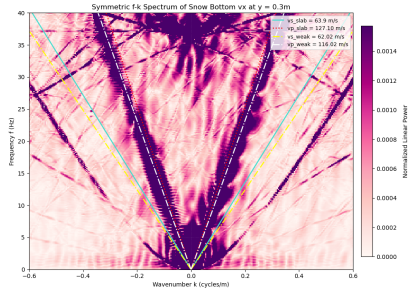
**Figure:** MPM-guided model velocity plots. Crack depth on left side, snow bottom on right.  $v_y$  at top,  $v_x$  at bottom. Velocity derived from displacement.

# Simple model F-k plots and dispersion curves



**Figure:** Supershear and Sub-Rayleigh F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

# MPM F-k plots



**Figure:** MPM F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

# MPM STF cumulative Integral

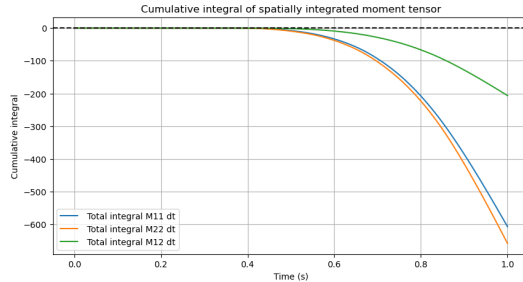


Figure: Cumulative integral over time of all MPM STF components





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